Alert Management Practice:

1. Introduction:

Alert management is a critical function in a Security Operations Center (SOC). It involves monitoring, classifying, prioritizing, and responding to security alerts generated by security tools such as SIEM, EDR, and IDS/IPS. Effective alert management helps analysts quickly identify real threats, reduce false positives, and respond to incidents efficiently.

In this task, alert management activities were simulated using **Google Sheets**, **Wazuh**, and **TheHive** to practice alert classification, prioritization, incident ticket creation, and escalation procedures.

2. Objectives:

The main objectives of this exercise were:

- To understand the alert handling process in a SOC environment.
- To classify alerts based on severity and threat type.
- To map alerts to the **MITRE ATT&CK** framework.
- To prioritize alerts using **Common Vulnerability Scoring System (CVSS)** scores.
- To document incidents and escalate critical alerts.

3. Tools Used:

3.1 Google Sheets

Used to create an alert classification table and prioritize alerts based on severity and CVSS score.

3.2 Wazuh

An open-source SIEM platform used to monitor and visualize alerts through dashboards.

3.3 The Hive

A security incident response platform used for creating and managing incident tickets.

4. Alert Classification System:

An alert classification system was created using **Google Sheets** to categorize alerts based on type, priority, and associated threat techniques.

Sample Alert Classification Table

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	Brute Force Login	Medium	T1110
003	Malware Execution	Critical	T1059
004	Port Scan	Low	T1046

This classification helps analysts quickly understand the nature of alerts and map them to known adversary tactics.

5. Alert Prioritization

Alerts were prioritized based on severity and vulnerability scoring using **CVSS**.

Example simulated alerts:

Alert	Description	CVSS Score	Priority
Log4Shell Exploit Attempt	Remote code execution vulnerability	9.8	Critical
Phishing Email	Suspicious link detected	8.0	High
Failed Login Attempts	Possible brute force attack	6.5	Medium
Port Scan	Network reconnaissance	3.7	Low

Higher CVSS scores indicate higher risk and require faster response.

6. Dashboard Visualization:

Using **Wazuh**, a dashboard was configured to visualize alert severity levels.

The dashboard displayed:

- Distribution of alerts (Critical, High, medium, and Low)
- Top attack sources
- Most triggered detection rules
- Alert trends over time

This helps SOC analysts quickly identify high-priority threats.

7. Incident Ticket Creation

When a critical alert is detected, an incident ticket is created in **TheHive**.

Example Incident Ticket

Title:
[Critical] Ransomware Detected on Server-X

Description:
Suspicious ransomware activity detected on Server-X. The file `crypto_locker.exe` was executed, and unusual outbound communication was detected from IP address `192.168.1.50`.

Indicators of Compromise (IOCs):

- Malicious File: `crypto_locker.exe`
- Suspicious IP: `192.168.1.50`
- Modified encrypted files detected.

8. Escalation Procedure

Critical alerts must be escalated to Tier-2 analysts for deeper investigation.

Escalation Steps

1. Verify the alert and confirm indicators.
2. Collect evidence such as logs and affected host details.
3. Contain the threat (e.g., isolate the system).
4. Create an incident ticket in the Hive.
5. Escalate the incident to Tier-2 SOC analysts.

Example escalation communication includes:

- Summary of incident.
- Indicators of compromise.
- Actions taken during initial triage.
- Required next steps.

9. Outcome

Through this exercise, the following skills were developed:

- Alert triage and prioritization.
- Incident documentation.
- Threat mapping using MITRE ATT&CK.
- Security monitoring using Wazuh dashboards.
- Incident management using Hive.

Response Documentation:

1. Introduction

Response documentation is an essential part of incident response in a Security Operations Center (SOC). It ensures that every security incident is properly recorded, investigated, and analyzed for future improvements. Proper documentation helps teams understand attack patterns, improve response procedures, and maintain compliance with security standards.

2. Objectives

The objectives of this exercise were:

- To learn how SOC teams document incident response activities.
- To create a standardized incident response report template.
- To record investigation actions in a structured format.
- To develop response checklists for phishing incidents.
- To conduct a post-mortem analysis to improve future response processes.

3. Tools Used

3.1 Google Docs

Used to create incident response templates, investigation documentation, and incident reports.

3.2 draw.io

Used to design incident response workflow diagrams and investigation flowcharts.

4. Incident Response Template

A standardized incident response template was created based on guidelines from the **SANS Institute** incident handling framework.

Example: Phishing Incident Report

1. Executive Summary

A phishing email containing a malicious link was detected and reported by a user. The message attempted to trick recipients into entering login credentials on a fake website. The email was analyzed and confirmed to be malicious.

2. Timeline

Time	Event
13:45	Phishing email received
13:55	The user reported email to SOC
14:00	The email was analyzed and confirmed malicious
14:10	Affected inboxes scanned

3. Impact Analysis

- Three users received the phishing email
- No credentials were compromised
- No malware was downloaded

4. Remediation Steps

- Blocked sender domain in email security gateway
- Removed phishing email from all mailboxes

- Updated spam filtering rules
- Conducted phishing awareness notification

5. Lessons Learned

- Users should report suspicious emails immediately
- Email filtering rules require improvement
- Security awareness training should be repeated

5. Investigation Steps Documentation

All actions taken during the incident investigation were logged in a structured table.

Example Investigation Log

Timestamp	Action
2025-08-18 14:00:00	Isolated affected endpoint
2025-08-18 14:10:00	Analyzed phishing email headers
2025-08-18 14:20:00	Checked malicious link reputation
2025-08-18 14:30:00	Collected memory dump
2025-08-18 14:45:00	Reviewed authentication logs

Maintaining this log ensures transparency and helps recreate the incident timeline.

6. Phishing Investigation Checklist:

A checklist was created in **Google Docs** to standardize phishing investigations.

Example Phishing Response Checklist

- Confirm email headers and sender domain.
- Analyze email attachments or links.
- Check URL reputation using **VirusTotal. Total.**
- Identify affected users.
- Search for similar emails in the organization.
- Block malicious sender or domain.
- Remove phishing emails from mailboxes.
- Notify affected users and provide guidance.

This checklist helps analysts respond quickly and ensures that no investigation steps are missed.

7. Post-Incident Review (Post-Mortem):

After handling the simulated incident, a short post-incident review was conducted.

Example Post-Mortem:

The phishing simulation highlighted the need for faster detection and improved email filtering. Although no accounts were compromised, several users received the malicious email before it was blocked. Future improvements include enhanced email security rules, faster SOC alerting, and regular employee phishing awareness training to reduce risk.

8. Outcome:

This exercise helped develop practical SOC documentation skills, including:

- Incident reporting and documentation
- Structured investigation logging
- Phishing incident response procedures
- Post-incident analysis and improvement planning

Alert Triage Practice:

1. Introduction

Alert triage is a critical process in a Security Operations Center (SOC) where analysts evaluate security alerts to determine their validity, severity, and potential impact. The main objective of alert triage is to quickly identify true security threats while filtering out false positives. This process helps SOC teams prioritize incidents and respond effectively to potential attacks.

2. Objectives

The objectives of this task were:

- To understand the process of alert triage in a SOC environment.
- To analyze security alerts generated by monitoring tools.
- To identify false positives and real threats.
- To validate indicators of compromise (IOCs) using threat intelligence platforms.

3. Tools Used

3.1 Wazuh

An open-source SIEM and security monitoring platform used to detect and analyze security alerts.

3.2 VirusTotal

A threat intelligence platform used to analyze suspicious files, domains, URLs, and IP addresses.

3.3 AlienVault OTX

A community-driven threat intelligence platform used to identify known malicious indicators and attack patterns.

4. Triage Simulation

A simulated alert was analyzed using Wazuh to determine whether it represented a legitimate threat.

Sample Alert Analysis

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	192.168.100.16	Medium	Open

Alert Investigation

The alert indicated multiple failed SSH login attempts from the IP address **192.168.100.16**. This behavior is commonly associated with brute-force attacks where attackers repeatedly attempt different username and password combinations to gain unauthorized access.

The investigation involved reviewing authentication logs, checking the number of failed login attempts, and verifying whether the IP address belonged to an internal system or an external source.

5. Threat Intelligence Validation

The suspicious IP address was analyzed using **VirusTotal** and **AlienVault OTX** to determine if it was associated with known malicious activity.

IOC Validation Steps

1. Extract the source IP address from the alert.
2. Search the IP address in VirusTotal.
3. Cross-reference the IP in AlienVault OTX.
4. Check reputation scores and related threat reports.

6. Threat Intelligence Summary

The IP address associated with the SSH brute-force alert was analyzed using threat intelligence platforms. The results showed that the IP had previous reports related to credential brute-force attempts. Based on the investigation, the alert was classified as a potential attack attempt, and monitoring and blocking actions were recommended.

7. Outcome

- Security alert investigation and analysis.
- Identification of brute-force attack patterns.
- Threat intelligence validation using IOC databases.
- Differentiating between false positives and real threats.
- Prioritizing alerts based on severity and impact.

Evidence Preservation:

1. Introduction

Evidence preservation is a critical step in digital forensics and incident response. During a security investigation, it is essential to collect and preserve system evidence in a way that maintains its integrity and ensures it can be used for further analysis or legal proceedings. Proper evidence preservation helps investigators reconstruct attack events and identify the root cause of incidents.

In this exercise, evidence preservation techniques were practiced using **Velociraptor** and **FTK Imager** to collect volatile data, create memory dumps, and document evidence using a chain-of-custody record.

2. Objectives

The objectives of this task were:

- To understand the importance of evidence preservation in incident response
- To collect volatile system data during a simulated security incident
- To perform memory acquisition for forensic analysis
- To document evidence using chain-of-custody procedures
- To verify the integrity of collected evidence using cryptographic hashes

3. Tools Used

3.1 Velociraptor

An open-source digital forensics and incident response (DFIR) tool used to collect system artifacts and perform remote investigations.

3.2 FTK Imager

A forensic acquisition tool used to create forensic disk and memory images without modifying the original data.

4. Volatile Data Collection

Volatile data refers to information stored in system memory that can be lost when a system is powered off. Examples include active processes, network connections, and system sessions. During the exercise, volatile data was collected from a Windows virtual machine using Velociraptor.

Command Used

SELECT * FROM netstat

This query collected active network connection details such as

- Local and remote IP addresses
- Open ports
- Network connection states

The results were exported and saved as a **CSV file** for further analysis and documentation.

5. Memory Evidence Collection

Memory acquisition was performed to capture the complete RAM contents of the system for forensic investigation. Memory analysis can reveal:

- Running processes
- Malware artifacts
- Encryption keys
- Suspicious network connections

Memory Acquisition Query

SELECT * FROM Artifact. Windows. Memory. Acquisition

The captured memory dump was then hashed using the **SHA-256** hashing algorithm to ensure the integrity of the evidence.

Hashing Command

sha256sum memory_dump. raw

Hash values ensure that the evidence has not been altered after collection.

6. Chain-of-Custody Documentation

A chain-of-custody document was created to record how digital evidence was collected, handled, and stored. This ensures accountability and preserves the integrity of the evidence.

Example Evidence Record

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Memory Dump	SOC Analyst	2025-08-18	`<SHA256>`
Network Connections	Netstat Output CSV	SOC Analyst	2025-08-18	`<SHA256>`

The documentation records the details of the evidence, the individual who collected it, and the cryptographic hash used to verify its integrity.

7. Importance of Evidence Preservation

Proper evidence preservation ensures:

- Data integrity during forensic investigations
- Accurate reconstruction of attack events
- Compliance with digital forensic standards
- Admissibility of evidence in legal proceedings

By preserving volatile and non-volatile evidence correctly, investigators can perform deeper forensic analysis without compromising the original data.

8. Outcome

- Digital evidence collection techniques.
- Volatile memory acquisition.

- Network artifact collection.
- Evidence integrity verification using SHA-256 hashes.
- Proper chain-of-custody documentation.

This activity provided practical experience in digital forensics procedures commonly used in SOC and incident response investigations.

Capstone Project: Full Alert-to-Response Cycle

1. Introduction

The capstone project simulated a real-world Security Operations Center (SOC) workflow, covering the entire process from attack detection to incident response and reporting. The goal was to practice how security analysts detect attacks, investigate alerts, contain threats, and document the incident.

The exercise used **Metasploit** to simulate an attack on **Metasploitable2**, while **Wazuh** monitored the system for security alerts. After detecting the attack, the response process involved blocking the attacker using **CrowdSec** and documenting the incident in **Google Docs**.

2. Objectives

The objectives of this capstone project were:

- Simulate a cyberattack in a controlled lab environment
- Detect the attack using a SIEM system
- Perform alert triage and incident investigation
- Apply containment measures to stop the attack
- Document the full incident response process

3. Attack Simulation

The attack simulation targeted a vulnerable FTP service on the Metasploitable2 virtual machine. The exploit used was the **vsftpd 2.3.4 backdoor vulnerability**.

Exploit Module

use exploit/unix/ftp/vsftpd_234_backdoor

The attacker machine initiated the exploit against the vulnerable FTP service, which resulted in unauthorized shell access to the target system.

This simulation helped demonstrate how attackers exploit known vulnerabilities to gain system access.

4. Detection and Alert Triage

The attack activity was detected by Wazuh through monitoring of system logs and network events.

Alert Log

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 11:00:00	192.168.100.16	VSFTPD exploit detected	T1190

The alert corresponds to the **MITRE ATT&CK** technique **Exploit Public-Facing Application (T1190)**.

During the triage process, the SOC analyst reviewed the logs, verified the source IP address, and confirmed that the activity matched the characteristics of an exploit attempt.

5. Incident Response

After confirming the attack, the following response actions were taken:

1. The compromised virtual machine was isolated from the network.

2. The attacker's IP address was blocked using CrowdSec.

3. System logs and forensic artifacts were preserved for analysis.

4. Network connectivity from the attacker was tested to confirm blocking.

Verification

A ping test from the attacker machine confirmed that the target system was no longer reachable, indicating that containment was successful.

6. Incident Report (200 Words)

Executive Summary

A simulated cyber attack targeting a vulnerable FTP service was successfully detected and mitigated during the SOC capstone exercise. The attack exploited the vsftpd 2.3.4 backdoor vulnerability on a Metasploitable2 virtual machine using Metasploit. The security monitoring system identified suspicious activity and generated alerts indicating a potential exploitation attempt.

Timeline

At 11:00 AM, the attack was launched from IP address 192.168.1.100. Shortly after, Wazuh generated an alert indicating a possible VSFTPD exploitation attempt. The SOC analyst reviewed the alert, confirmed the suspicious activity, and initiated the incident response procedure.

Response

The affected system was immediately isolated from the network to prevent further compromise. The attacker's IP address was blocked using CrowdSec to stop additional malicious activity. System logs and evidence were preserved for investigation.

Recommendations

To prevent similar attacks in the future, it is recommended to regularly patch vulnerable services, disable unnecessary services such as outdated FTP servers, implement stronger network monitoring, and conduct regular vulnerability assessments.

7. Stakeholder Briefing

A simulated security incident was detected during the SOC monitoring exercise involving an attempted exploitation of a vulnerable FTP service on one of our systems. The attack originated from an internal testing machine and was quickly identified by our monitoring tools. Our security team investigated the alert, confirmed the activity, and immediately isolated the affected system to prevent further access. The attacker's IP address was blocked, and system logs were preserved for analysis. No production systems or sensitive data were impacted during this exercise. The incident highlights the effectiveness of our monitoring controls and emphasizes the importance of regular vulnerability patching and security monitoring.

8. Outcome

This capstone project demonstrated the complete SOC workflow, including

- Attack simulation and vulnerability exploitation.
- SIEM alert detection and analysis.
- Threat triage and investigation.
- Incident containment and response.
- Security incident documentation and reporting.